Bounding a trained network's error by compressing it, without validation data

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Abstract

If a trained network f agrees on most inputs with a network g that has a short description, then an Occam bound on g plus the disagreement $\Pr[f \neq g]$ bounds the test error of f, with no validation data. We study when the disagreement term can be certified. When the input distribution is known, as for Boolean tasks with uniform inputs, it is free. A 289,281-parameter MLP then compresses into a 28–100-bit truth table, threshold function or small quantized network, and the bound tracks the test error of f along the whole learning curve. This includes networks that have only partly learned a parity and networks that interpolate 10% label noise, where parameter-counting and VC bounds are vacuous. On 20-bit inputs the compressed networks agree with f everywhere, giving bounds valid for every input distribution. With only the training set, we prove a dithered transfer theorem. Compressing f by subtractive dithered quantization with public randomness makes $g - f$ independent noise. A certificate computed from g alone at each training point then transfers through the same Occam bound, giving a deterministic bound on f itself. This removes the $\sqrt{d}$ loss of worst-case perturbation sets. On MNIST it is non-vacuous for narrow ReLU networks and for networks with sparse updates, and vacuous for wide dense ones, because it needs the weights described to the precision the margins require. The random initialization costs nothing, since codes may depend on data-independent randomness. We measure how coding relative to the initialization, the initialization scale and a deterministic initialization change the description length, and give an init-kernel codec whose cost reproduces the n^k sample complexity of kernel methods.

1 Introduction

A standard way to certify that a trained network generalizes is to hold out labelled data. Without held-out data, the classical tools (VC dimension, parameter counting, norm-based bounds) are vacuous for networks with more parameters than training examples. They bound every network in a huge class, not the one that was trained.

This paper studies a reduction that targets the trained network itself. Suppose that some network g with a *short description* (a code of L bits fixed before seeing the data) agrees with the trained network f on most inputs. Then

$$\operatorname{err}(f) \;\leq\; \operatorname{err}(g) + \Pr_x[f(x) \neq g(x)],$$

and since g ranges over at most 2^L descriptions, a Chernoff–Occam union bound over all of them bounds $\operatorname{err}(g)$ by its training error plus about $L \ln 2/m$. Bounding generalization reduces to

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compressing the network, provided one can certify the closeness term $\Pr[f \neq g]$. That term concerns the input distribution, which is exactly what the training data does not reveal without depending on f.

We consider three regimes, in decreasing order of what is known about the input distribution, and ask three questions. Has this been done? Can a compression scheme give interesting bounds on real networks, beating parameter counting? Does the random initialization have to be described, which would make the description long?

Answers in brief.

- *Prior work.* Compression bounds for the compressed network g are well developed [1, 24, 41]. The reduction to the original f with the closeness term estimated on *unlabeled* data goes back to Kääriäinen [20] and was recently carried out for modern networks by Bazinet et al. [3], with non-vacuous bounds on MNIST and CIFAR-10. With no extra data, the deterministic derandomization bounds of Neyshabur et al. [31] and Nagarajan and Kolter [29] bound the original network but are vacuous. The exception we know of is Biggs and Guedj [5], who obtain non-vacuous training-data-only bounds for one-hidden-layer networks with erf or GELU activations, via an exact identity between such a network and a stochastic majority vote.

- *Known input distribution* (Sections 5.1 and 5.2). On Boolean tasks with uniform inputs, $\Pr[f \neq g]$ is free. A 289,281-parameter MLP trained on $n = 50$ bits compresses into a 28–100-bit truth table, linear threshold function or small quantized network. The resulting bound on f tracks its test error along the whole learning curve: within 0.005 at $m = 16000$, 0.30 against 0.29 for a partially trained parity network, and 0.17 against 0.14 for a network that interpolates 10% label noise. Parameter-counting and VC bounds are 1.0 throughout. For $n = 20$ the compressed networks agree with f on all 2^{20} inputs, giving bounds that hold for *every* input distribution.

- *Training data only* (Sections 3.1 and 5.4). We prove a *dithered transfer theorem.* Compress f by subtractive dithered quantization with public randomness, so that $g - f$ is independent uniform noise; compute from g alone a per-input certificate that f and g classify alike; and transfer the fraction of certified training points to the population with the same Occam bound. The result is a deterministic bound on the original f with no validation or unlabeled data. It is non-vacuous for narrow one-hidden-layer MNIST networks (0.44 for 8 hidden units) and for networks whose first-layer update is sparse in a DCT basis (Table 4), and it becomes vacuous for wide dense networks. The certificate needs f's weights described to the precision its margins require, and that costs bits in proportion to the width. Worst-case versions of the same idea fail by a factor $\sqrt{d}$ (Theorem 4).

- *Initialization* (Sections 3.2 and 5.5). The initialization costs nothing: any data-independent randomness may be used by the code (Lemma 9). Function-space codecs never refer to it, and weight-space codecs code the update relative to it. We measure how coding relative to the initialization or to zero, the initialization scale and a deterministic initialization affect the description length. We give an init-kernel codec (Proposition 10) whose cost reproduces the n^k sample complexity of kernel methods. In that picture, feature learning is what makes trained networks short to describe.

Scope. Everything here is a theorem or a reproducible measurement on small networks: MLPs on 20–50-bit Boolean tasks, MNIST and Fashion-MNIST, on a 4-core CPU. Every bound is computed

with explicit code lengths, confidence parameters and a charge for the design choices we made on the same data (Section 5). The code, the result files and an AI referee record are in the repository.

2 Related work

Occam, MDL and PAC-Bayes. Bounding the error of a hypothesis by its description length is the Occam bound [7, 21]. Applied to network weights it is the minimum-description-length programme of Hinton and van Camp [16] and the flat-minima argument of Hochreiter and Schmidhuber [17]. PAC-Bayes [26] lets the hypothesis be a distribution over weights; Langford and Caruana [22] and Dziugaite and Roy [11] obtained non-vacuous PAC-Bayes bounds for stochastic networks, the latter with a prior centred at the random initialization, and Pérez-Ortiz et al. [32] and Dziugaite et al. [12] tightened them with data-dependent priors.

Bounds for compressed networks. Arora et al. [1] bound a compressed network g obtained by random projections from a trained f. Zhou et al. [41] combine pruning, quantization and coding with a noisy posterior and obtain non-vacuous bounds at ImageNet scale for the noisy compressed network. Lotfi et al. [24] train in a random subspace [23], quantize and arithmetic-code, and obtain the tightest data-independent bounds we know of (11.6% on MNIST and 32.8% on Fashion-MNIST, values from their repository); Lotfi et al. [25] extend this to language models. In all of these the certified predictor is the compressed (or noisy, or subspace-trained) network, not an independently trained f.

Bounds for the original network through a surrogate. The reduction $\mathrm{err}(f) \le \mathrm{err}(g) + \Pr[f \neq g]$ with the disagreement estimated on unlabeled data goes back at least to Kääriäinen [20]. Hsu et al. [18] use a distilled network and control the distance with samples from an augmentation distribution. Garg et al. [13] certify the original network using randomly labelled unlabeled data. Most directly, Bazinet et al. [3] implement the reduction for modern networks with surrogates certified by sample compression, PAC-Bayes or Lotfi-style compression, and disagreement on a held-out unlabeled set. Their repository reports bounds on the unmodified network of 1.78–3.45% for an MNIST CNN, depending on the surrogate, and 17–35% for a CIFAR-10 ResNet-18. Our unlabeled-data experiments are an instance of their method. Theoretical treatments of the same step without extra data include Suzuki et al. [35] (local Rademacher complexity of the difference class), Daniely and Granot [10] (approximation at every input), Sefidgaran et al. [34] (rate-distortion) and the surrogate framework of Negrea et al. [30]; none yields numerical bounds.

Deterministic bounds from noise resilience. Neyshabur et al. [31] and Nagarajan and Kolter [29] derandomize PAC-Bayes for the original network using margins and robustness to weight perturbations, checked on the training data; Wei and Ma [37] use an all-layer margin. These bounds are vacuous in practice. Biggs and Guedj [5] obtain non-vacuous bounds from the training data for deterministic one-hidden-layer networks with erf or GELU activations (about 0.7–0.8 on MNIST without a data-dependent prior, by our reading of their slides; unverified). Their bounds rest on the exact identity $\mathbb{E}[\mathrm{sign}(w \cdot x + \mathrm{noise})] \propto \mathrm{erf}(\cdot)$, which makes such a network a PAC-Bayes majority vote; see also Biggs and Guedj [4]. Our dithered transfer theorem shares their structure (a perturbation set around f plus a per-point condition transferred from training to test), but it uses a single public random draw, an Occam code of the perturbed network, and a per-input concentration bound in place of worst-case or norm-based perturbation bounds. Sound bound propagation under weight

perturbations [38] and subtractive dither [33] are the tools; shared-randomness model coding also appears in Havasi et al. [15], and single-draw arguments in Blanchard and Fleuret [6].

Boolean tasks and simplicity. Valle-Pérez et al. [36] bound the Bayesian posterior of networks on 7-bit Boolean functions with a prior induced by random initialization. Our Boolean experiments instead bound the network produced by gradient descent, via distillation into a junta, a linear threshold function or a small quantized network, with the disagreement computed exactly or on free samples from the known input distribution. A companion project in the same repository [9] reports similar certificates for rich-regime networks on Boolean targets, using unlabelled inputs. When the labelling function is also known, a network's error can be computed outright; Gross et al. [14] study proofs of such statements and the trade-off between proof length and tightness.

Initialization. A data-independent initialization costs nothing in PAC-Bayes [11] and in Occam codes [1, 24]. Supermasks [40] show that a random initialization can carry most of the information of a good network at zero cost. Lazy and rich training [8, 19, 27] predict how the update $\theta - \theta_0$ is structured, and deterministic initializations exist [39]. We are not aware of a study measuring the description length of the update as a function of the initialization scale, or comparing coding relative to the initialization and relative to zero.

3 From a short description to a bound on the trained network

Setting. D is a distribution on $\mathcal{X} \times \mathcal{Y}$ with input marginal D_X, and the training sample is $S = ((x_i, y_i))_{i \le m} \sim D^m$. For a predictor h, $\mathrm{err}(h) = \Pr_D[h(x) \neq y]$ and $k_S(h) = \#\{i : h(x_i) \neq y_i\}$. The trained network $f = f_S$ is an arbitrary function of S and of any other randomness used in training. A *code* is a countable set Σ of descriptions with lengths $L : \Sigma \to [0, \infty)$ satisfying $\sum_\sigma 2^{-L(\sigma)} \le 1$, fixed before S is seen; σ decodes to a predictor g_σ. All our lengths are $-\log_2$ of products of conditional probabilities (Elias-δ codes for integers, enumerative codes for subsets, adaptive Krichevsky–Trofimov codes for integer arrays), so they need not be integers. We write $\overline{\mathrm{Bin}}(k, m, \alpha) = \sup\{p : \Pr[\mathrm{Bin}(m, p) \le k] \ge \alpha\}$ for the binomial tail inversion.

Lemma 1 (Occam bound, exact binomial form)**.** *With probability at least $1-\delta$ over S, simultaneously for all $\sigma \in \Sigma$,*

$$\mathrm{err}(g_\sigma) \le \overline{\mathrm{Bin}}(k_S(g_\sigma), m, \delta 2^{-L(\sigma)}) \le \mathrm{kl}^{-1}\Big(\tfrac{k_S(g_\sigma)}{m},\ \tfrac{L(\sigma)\ln 2 + \ln(1/\delta)}{m}\Big).$$

Proof. For fixed σ, $k_S(g_\sigma) \sim \mathrm{Bin}(m, \mathrm{err}(g_\sigma))$, so $\mathrm{err}(g_\sigma) > \overline{\mathrm{Bin}}(k_S, m, \alpha)$ says that k_S falls in a lower tail of mass less than $\alpha = \delta 2^{-L(\sigma)}$. Sum over σ using Kraft. The second inequality is the Chernoff bound $\Pr[\mathrm{Bin}(m, p) \le qm] \le e^{-m\,\mathrm{kl}(q\|p)}$ for $q \le p$. □

The description may be chosen after seeing S in any way.

Theorem 2 (reduction to compression)**.** *For any predictor f and any description $\sigma = \sigma(S)$, $\mathrm{err}(f) \le \mathrm{err}(g_\sigma) + \Pr_{D_X}[f(x) \neq g_\sigma(x)]$. If, after σ is fixed, N inputs are drawn i.i.d. from D_X independently of everything else and $f \neq g_\sigma$ on j of them, then with probability at least $1 - \delta_1 - \delta_2$,*

$$\mathrm{err}(f) \le \overline{\mathrm{Bin}}(k_S(g_\sigma), m, \delta_1 2^{-L(\sigma)}) + \overline{\mathrm{Bin}}(j, N, \delta_2).$$

Proof. $\{f(x) \neq y\} \subseteq \{g(x) \neq y\} \cup \{f(x) \neq g(x)\}$. The first term is Lemma 1; conditional on (S, σ) the disagreement indicators are i.i.d. Bernoulli with mean $\Pr_{D_X}[f \neq g_\sigma]$, so Clopper–Pearson gives the second. □

The disagreement term needs *inputs* from D_X but no labels. We use three sources: (i) a known, sampleable D_X (Boolean tasks with uniform inputs: the inputs are free, and for $\{\pm 1\}^n$ with $n \le 22$ we enumerate them, making the term exact); (ii) unlabeled samples (MNIST test images without labels; this is the setting of Bazinet et al. [3]); (iii) none at all (Section 3.1). If K candidate surrogates are compared on the same N inputs, δ_2 is replaced by δ_2/K; our experiments either select on a separate pool or apply this correction.

Corollary 3 (distribution-free, finite domain)**.** *Let $\mathcal{X}$ be finite. If a description g' satisfies $g'(x) = f(x)$ for all $x \in \mathcal{X}$, then for every distribution D, with probability $\ge 1-\delta$, $\mathrm{err}_D(f) \le \overline{\mathrm{Bin}}(k_S(f), m, \delta 2^{-L(g')})$. A short g that disagrees with f on a set E becomes such a g' by appending E, at a cost of $1 + L_\delta(|E|+1) + \log_2 \binom{|\mathcal{X}|}{|E|}$ bits.*

3.1 Bounds from training data alone

The disagreement term is a statement about D_X. To avoid it, make the closeness certificate a function of the short description only, evaluate it at the training inputs, and transfer it with the same Occam bound.

Theorem 4 (certified transfer)**.** *Let each σ define a set K_σ of predictors and a measurable $\pi_\sigma : \mathcal{X}\times\mathcal{Y} \to \{0,1\}$ with $\pi_\sigma(x,y) = 1 \Rightarrow h(x) = y$ for all $h \in K_\sigma$. With probability $\ge 1-\delta$, simultaneously for all σ and all $h \in K_\sigma$,*

$$\mathrm{err}(h) \le \overline{\mathrm{Bin}}(\#\{i : \pi_\sigma(x_i, y_i) = 0\}, m, \delta 2^{-L(\sigma)}).$$

Proof. $\mathrm{err}(h) \le \Pr_D[\pi_\sigma = 0]$ for $h \in K_\sigma$; apply Lemma 1 to the losses $1 - \pi_\sigma$. □

For networks, K_σ is a set of weights around a quantized network and π_σ a sound interval bound; $f \in K_\sigma$ is checked on the weights. This works poorly for dense networks. If the residual $f - g$ is spread over d coordinates, the worst case of $|\Delta w \cdot x|$ over a box or ball containing it (any set that is cheap to describe without knowing the residual's direction relative to x) is larger than its actual value by a factor of order $\sqrt{d}$. On a 784–100–10 MNIST network the ratio of certified to actual first-layer error is 12–13 at every step size we tried, and certifying 95% of the training points needs about 8 bits per first-layer coefficient (Section 5.4). Randomizing the quantization removes the $\sqrt{d}$.

Dithered compression. Let ξ be public randomness drawn independently of S, of the test point and of any auxiliary randomness ζ used to tune the encoder. In our code ξ is generated from fresh operating-system entropy after training and after tuning, and the pseudo-random generator is modelled as truly random, as for any randomized bound. An encoder maps (f, ξ) to a description $\sigma = \sigma_{S,\zeta}(f, \xi)$; it may depend on S and ζ arbitrarily, for instance through step sizes chosen on the training data. Each (σ, ξ) defines a certificate $\pi_{\sigma,\xi}$.

Theorem 5 (dithered transfer)**.** *Suppose that for every fixed encoder (every value of (S,ζ)), every network f and every (x,y) there is an event $\mathrm{Bad}_{x,y}(f,\xi)$ with $\Pr_\xi[\mathrm{Bad}_{x,y}] \le \eta$ such that $\pi_{\sigma(f,\xi),\xi}(x,y) = 1$ and not $\mathrm{Bad}_{x,y}$ imply $f(x) = y$. Then with probability $\ge 1 - \delta - \delta'$ over (S,ξ),*

$$\mathrm{err}(f) \le \overline{\mathrm{Bin}}(\#\{i : \pi(x_i, y_i) = 0\}, m, \delta 2^{-L(\sigma)}) + \eta/\delta'.$$

Proof. $\mathrm{err}(f) \le \Pr_D[\pi = 0] + \Pr_D[\mathrm{Bad}]$. Conditional on ξ, S is i.i.d. from D and $\{\pi_{\sigma,\xi}\}_\sigma$ is a fixed family, so Lemma 1 bounds the first term for all σ at once. Conditional on (S,ζ), which fixes f and the encoder, Fubini gives $\mathbb{E}_\xi \Pr_D[\mathrm{Bad}] = \mathbb{E}_D \Pr_\xi[\mathrm{Bad}] \le \eta$, and Markov's inequality gives $\Pr_\xi[\Pr_D[\mathrm{Bad}] > \eta/\delta'] \le \delta'$. □

Hyperparameters chosen by looking at ξ itself would need a union bound in the Markov step; we choose them with an independent selection dither ζ and certify with a fresh ξ. Hyperparameters that change the certificate (step sizes, the basis, t_1, t_2) must be part of σ or fixed in advance, so that the Occam step covers them.

Lemma 6 (subtractive dither [33])**.** *For $a \in \mathbb{R}$, $s > 0$ and $U \sim \mathrm{Unif}[-\frac{1}{2}, \frac{1}{2}]$, the error $s(\mathrm{round}(a/s + U) - U) - a$ is uniform on $[-\frac{s}{2}, \frac{s}{2}]$ whatever a is. With independent U's per coordinate, the errors are independent, the error in a coordinate with step s_k is uniform on $[-\frac{s_k}{2}, \frac{s_k}{2}]$, and their joint law does not depend on f.*

The decoder knows ξ and the integers $K = \mathrm{round}(a/s + \xi)$, so g costs $L(K)$ bits while $g - f$ is noise of known law. The next lemma turns this into a per-input certificate for one-hidden-layer networks.

Lemma 7 (certificate for one-hidden-layer networks)**.** *Let $f(x) = V\mathrm{relu}(Wx + b) + c$ with $W = W_0 + AB^\top$ (W_0 the initialization, B an orthonormal basis), $V \in \mathbb{R}^{C \times H}$, and let g be obtained by subtractive dithering of A (steps $s_{ik} \geq 0$; a step of 0 means the coefficient is sent exactly, as we do for coefficients that are exactly zero), $b - b_0$ (step s^b), $V - V_0$ (s^V) and $c - c_0$ (s^c). Fix (x, y), put $\hat{x} = B^\top x$, $v_i = \sum_k s_{ik}^2 \hat{x}_k^2/12 + (s^b)^2/12$ and $\tau_i = t_1\sqrt{v_i}$, and let u'_i, z' be g's pre-activations and logits. For $c \neq y$ let $D_i = |V'_{yi} - V'_{ci}| + s^V$, let w_i be 1 if $u'_i \geq 2\tau_i$, 0 if $u'_i \leq -2\tau_i$ and 2 otherwise, and*

$$\hat{P}_c = \textstyle\sum_i w_i D_i^2 v_i + \frac{(s^V)^2}{6} \sum_i \mathrm{relu}(u'_i + 2\tau_i)^2 + \frac{(s^c)^2}{6}, \qquad \hat{B}_c = \textstyle\sum_{i:|u'_i|<2\tau_i} D_i \hat{\beta}_i,$$
$$\hat{\beta}_i = \min\Big(\tfrac{\sqrt{v_i}}{2},\ \sqrt{2\pi v_i}\,\bar{\Phi}\big((|u'_i| - \tau_i)_+/\sqrt{v_i}\big)\Big).$$

Let $\pi_g(x, y) = 1$ iff $z'_y - z'_c > t_2\sqrt{2\hat{P}_c} + \hat{B}_c$ for every $c \neq y$. Then the hypothesis of Theorem 5 holds with $\eta = 2He^{-t_1^2/2} + 2(C-1)e^{-t_2^2}$. (Binary networks with one logit: $D_i = |V'_i| + s^V/2$, the last two terms of $\hat{P}$ carry $1/12$, and $C - 1$ is replaced by 1.)

Proof. Let ϵ, e, Ξ, γ be the dither errors of A, b, V, c, $\eta_i = \sum_k \epsilon_{ik}\hat{x}_k + e_i$, and u_i the pre-activations of f, so $u'_i = u_i + \eta_i$.

(a) Tails. If $U \sim \mathrm{Unif}[-a, a]$ then $\mathbb{E}e^{\lambda U} = \sinh(\lambda a)/(\lambda a) \leq e^{\lambda^2 a^2/6}$, so U is sub-Gaussian with variance proxy $a^2/3$, its variance. Hence the η_i are independent, symmetric and sub-Gaussian with proxies v_i, and $G = \{|\eta_i| \leq \tau_i\ \forall i\}$ has $\Pr[G^c] \leq 2He^{-t_1^2/2}$.

(b) Decomposition. Fix $c \neq y$; let $d_i = V_{yi} - V_{ci}$ and $\chi_i = \Xi_{yi} - \Xi_{ci}$. The margin error is $F = \sum_i d_i(\mathrm{relu}(u_i + \eta_i) - \mathrm{relu}(u_i)) + \sum_i \chi_i \mathrm{relu}(u_i + \eta_i) + \gamma_y - \gamma_c$. Replacing each η_i by $\mathrm{clip}_{\tau_i}(\eta_i)$ gives $\tilde{F}$, and $F = \tilde{F}$ on G. With $\varphi_i(t) = \mathrm{relu}(u_i + \mathrm{clip}_{\tau_i}(t)) - \mathrm{relu}(u_i)$: if $u_i \geq \tau_i$ then $\varphi_i = \mathrm{clip}_{\tau_i}$ (*active*); if $u_i \leq -\tau_i$ then $\varphi_i \equiv 0$ (*inactive*); otherwise φ_i is 1-Lipschitz (*marginal*).

(c) Moment generating function. Conditional on $(\eta_i)_i$, the Ξ-term is a centred linear form in independent uniforms with proxy at most $\frac{(s^V)^2}{6}\sum_i \mathrm{relu}(u_i + \tau_i)^2$, and $\gamma_y - \gamma_c$ has proxy $(s^c)^2/6$. For an active unit, $\mathrm{clip}(\eta_i)$ is symmetric with $|\mathrm{clip}(\eta_i)| \leq |\eta_i|$, so $\mathbb{E}e^{\lambda d_i \mathrm{clip}(\eta_i)} = \mathbb{E}\cosh(\lambda d_i \mathrm{clip}(\eta_i)) \leq \mathbb{E}\cosh(\lambda d_i \eta_i) \leq e^{\lambda^2 d_i^2 v_i/2}$. For a marginal unit, let X' be an independent copy of $X = \eta_i$; by Jensen, $\mathbb{E}e^{\lambda(\varphi(X) - \mathbb{E}\varphi(X))} \leq \mathbb{E}e^{\lambda(\varphi(X) - \varphi(X'))} = \mathbb{E}\cosh(\lambda(\varphi(X) - \varphi(X'))) \leq \mathbb{E}\cosh(\lambda(X - X')) \leq e^{\lambda^2 \cdot 2v_i/2}$, a proxy of $2d_i^2 v_i$. Multiplying the independent factors, $\tilde{F} - \mathbb{E}\tilde{F}$ is sub-Gaussian with proxy $P = \sum_{\mathrm{act}} d_i^2 v_i + 2\sum_{\mathrm{marg}} d_i^2 v_i + \frac{(s^V)^2}{6}\sum_i \mathrm{relu}(u_i + \tau_i)^2 + \frac{(s^c)^2}{6}$, and $\Pr[|\tilde{F} - \mathbb{E}\tilde{F}| \geq t_2\sqrt{2P}] \leq 2e^{-t_2^2}$.

(d) Bias. Only marginal units contribute to $\mathbb{E}\tilde{F}$. For the symmetric variable $\tilde{\eta} = \mathrm{clip}(\eta)$, $0 \leq \mathbb{E}\mathrm{relu}(u + \tilde{\eta}) - \mathrm{relu}(u) = \mathbb{E}(\tilde{\eta} - |u|)_+$, which is at most $\mathbb{E}\tilde{\eta}_+ = \mathbb{E}|\tilde{\eta}|/2 \leq \sqrt{v}/2$ and at most $\mathbb{E}(\eta - |u|)_+ = \int_{|u|}^\infty \Pr[\eta > t]\,dt \leq \sqrt{2\pi v}\,\bar{\Phi}(|u|/\sqrt{v})$.

(e) Computing from g. On G, $|u_i - u'_i| \le \tau_i$: so $u'_i \ge 2\tau_i$ implies active, $u'_i \le -2\tau_i$ implies inactive, $|u_i| \ge (|u'_i| - \tau_i)_+$ and $\mathrm{relu}(u_i + \tau_i) \le \mathrm{relu}(u'_i + 2\tau_i)$; always $|d_i| \le |V'_{yi} - V'_{ci}| + s^V$. Hence on G, $P \le \hat{P}_c$ and $|\mathbb{E}\tilde{F}| \le \hat{B}_c$.

(f) Conclusion. Let $\mathrm{Bad} = G^c \cup \bigcup_{c \ne y}\{|\tilde{F}_c - \mathbb{E}\tilde{F}_c| \ge t_2\sqrt{2P_c}\}$, so $\Pr[\mathrm{Bad}] \le \eta$. Off Bad, $|F_c| < t_2\sqrt{2\hat{P}_c} + \hat{B}_c$ for all $c \ne y$, so $\pi_g(x, y) = 1$ forces $z_y > z_c$ for all $c \ne y$, i.e. $f(x) = y$. □

The certificate depends only on (K, steps, ξ) and (x, y), as Theorem 5 requires. The constants are what make the bound typical-case: the threshold scales as $t_2\sqrt{2\sum_i D_i^2 v_i}$, an ℓ_2 aggregate of the per-coordinate noise, instead of the ℓ_1 aggregate $\sum_i D_i \sum_k s_{ik}|\hat{x}_k|/2$ of a worst-case set.

Remark 8 (relation to PAC-Bayes). The law of $g = f + \epsilon$ is a posterior (uniform on a box around f), and $L(K)$ plays the role of its relative entropy to a prior. Theorem 5 uses one draw instead of an average over the posterior, pays η/δ' for it, and reaches the deterministic f through a per-input concentration statement. The margin-based derandomizations of Neyshabur et al. [31] and Nagarajan and Kolter [29] instead control the perturbation's effect uniformly over inputs, with norm-based bounds.

3.2 Initialization

Lemma 9 (the initialization is free)**.** *Let θ_0 be drawn from any distribution independent of S, and let the code $(\Sigma_{\theta_0}, L_{\theta_0})$ depend on θ_0 (for instance, it describes $\theta - \theta_0$). Then Lemma 1 holds for it with probability $\ge 1 - \delta$ over (S, θ_0).*

Proof. Conditional on $\theta_0 = t$, $S \sim D^m$; apply Lemma 1 to the fixed code (Σ_t, L_t) and average over t. □

The same holds for every other piece of data-independent randomness: random projections, the dither, the DCT basis. A seed chosen as the best of N after training costs $\log_2 N$ bits; we charge 32 bits for the seed throughout, although with the seed fixed in advance even that is unnecessary. What the initialization can still cost is indirect: the description of the update $\theta - \theta_0$ depends on how training used θ_0.

Proposition 10 (init-kernel codec, linearized regime)**.** *Let $J(x) = \partial_\theta f_{\theta_0}(x)$, $F_0 = \mathbb{E}_{D_X}[J(x)^\top J(x)]$, $f^{\mathrm{lin}}_{\theta_0+\Delta}(x) = f_{\theta_0}(x) + J(x)\Delta$ and P_r the projector onto the top-r eigenvectors of F_0. Then $\mathbb{E}_{D_X}\|f^{\mathrm{lin}}_{\theta_0+\Delta}(x) - f^{\mathrm{lin}}_{\theta_0+P_r\Delta}(x)\|^2 = \|F_0^{1/2}(I - P_r)\Delta\|^2 \le \lambda_{r+1}(F_0)\|(I - P_r)\Delta\|^2$.*

When D_X is known, F_0 depends only on θ_0 and D_X, so its eigenbasis is free (Lemma 9) and $\theta_0 + P_r\Delta$ is described by r quantized coefficients. Heuristically, for uniform inputs on $\{\pm 1\}^n$ and an initialization distribution invariant under permutations and sign flips of the input coordinates (e.g. i.i.d. symmetric weights), the *expected* kernel is a function of $\langle x, x' \rangle$; its eigenfunctions are Walsh characters, and the degree-ℓ eigenspace has dimension $\binom{n}{\ell}$. A lazily trained network fitting a degree-k target then needs $r \approx \sum_{\ell \le k}\binom{n}{\ell}$ coefficients, so this codec certifies generalization once $m \gtrsim n^k$, the familiar sample complexity of kernel methods on degree-k targets. A network that has found the k relevant coordinates is instead close to a k-junta, which costs $\log_2\binom{n}{k} + 2^k$ bits. In this description-length picture, feature learning is what makes trained networks short.

4 Compression schemes

All code lengths are computed by `code/cgb/coding.py`: integers by Elias-δ (signed integers by the zigzag map), subsets of $[n]$ of size k by $L_\delta(k+1) + \log_2\binom{n}{k}$, and integer arrays of known shape by

the cheaper of two codes plus one flag bit. The dense code sends the range and then an adaptive Krichevsky–Trofimov (KT) code over it; the sparse code sends the support as a subset and then the nonzero values with a KT code. Step sizes lie on the grid $2^{e/4}$, $e \in \mathbb{Z}$, and are sent as signed integers. A unit test checks the Kraft inequality for each component (`code/tests/test_bounds_coding.py`).

Boolean tasks: function-space compression. Inputs are uniform on $\{\pm 1\}^n$ and the trained network f is an MLP of widths $(n, 512, 512, 1)$. The codec only queries f on inputs drawn from the uniform distribution, independently of the training set. (i) It estimates the influences $\mathrm{Inf}_j(f) = \Pr[f(x) \neq f(x^{\oplus j})]$ and proposes coordinate sets J, the coordinates with influence above τ for $\tau \in \{0.3, \ldots, 0.005\}$, plus $J = [n]$. (ii) For each J with $|J| \leq 14$ it builds the *truth table* $g(x) = T[x_J]$, where T is the majority of f over each fibre (estimated on a pool of 10^5 uniform inputs) and is sent with a KT code. (iii) For each J and hidden width $h \in \{0, 1, 2, 4, 8, 16\}$ ($h = 0$ is a linear threshold function) it distils a student MLP on x_J from f's labels on the pool. It then quantizes the student at three step sizes with quantization-aware fine-tuning (straight-through estimator) and sends its integers. (iv) It keeps the candidate minimizing the estimated bound: the Occam term plus the disagreement on a separate selection pool of 3×10^4 uniform inputs. (v) The certified disagreement is computed afterwards, either exactly on all 2^n inputs ($n = 20$) or on 2×10^5 fresh inputs ($n = 50$). The description includes a 2-bit codec-family tag, the subset J, the architecture and the steps. None of the codec's choices needs a union bound in the Occam term, because they are all part of the description.

Images: weight-space compression relative to the initialization. For an MLP f on 28×28 images, the compressed g keeps f's architecture and initialization θ_0 (free by Lemma 9; we still charge 32 bits for the seed). It codes the update $\theta - \theta_0$, with the first layer's update expressed in the orthonormal 2-D DCT basis and quantized with one step per frequency band ($u + v \in [0, 4), [4, 8), [8, 14), [14, 22), [22, 32), [32, \infty)$). The integers are coded per band. Starting from f, g is fine-tuned through the quantizer, with either training labels, f's soft labels on the training inputs (distillation), or both. The objective adds a rate penalty $\lambda \sum |c|/s$ that pushes coefficients to zero. Both this fine-tuning and the choice of (step, λ, loss) may use the training set freely.

Images and Boolean tasks: dithered compression for training-data-only bounds. For one-hidden-layer networks, g is the subtractive dither of f (Lemma 6) with steps $s_{ik} = 2^{(e_i + e'_{\mathrm{band}(k)})/4}$. The per-band exponents follow $(\mathrm{rms}\, \hat{x}_{\mathrm{band}})^{-p_b}$ and the per-unit exponents follow $\|V_{:,i}\|^{-p_u}$, both times a global scale. The steps are part of the description, and the integers are coded per band. The biases and the output layer get a common step, a multiple $v_m \in \{1, 4, 16\}$ of 0.002 times the global scale. The global scale, the exponents (p_b, p_u) and v_m are chosen on a subsample of the training set with one dither draw; the bound is then computed with a fresh draw on the full training set. We use $t_1 = 5.8$ and $t_2 = 3.9$ throughout, fixed before the final runs and charged as one of four allowed pairs (2 bits); $\eta/\delta' \leq 2 \times 10^{-3}$ at $\delta' = 0.01$ for $H \leq 128$, and every reported bound uses the η of its own H. The description also carries 2 bits for the initialization scheme and the reference (initialization or zero).

Baselines. For every trained f we report the Occam bound with 32 bits per parameter (*parameter counting*) and the VC bound $\hat{e} + \sqrt{(d(\ln(2m/d) + 1) + \ln(4/\delta))/m}$ with d equal to the number of parameters. The latter is optimistic for the baseline, since the VC dimension of ReLU networks with W weights and L layers is $O(WL \log W)$ and $\Omega(WL \log(W/L))$ [2].

Confidence. $\delta = 0.05$ throughout. In Theorem 2 it is split equally between the two terms, unless the disagreement is computed exactly. For the dithered bounds, $\delta = 0.04$ and $\delta' = 0.01$.

5 Experiments

All numbers in this section are produced by the scripts in `code/experiments` and are reported in the summary files of `results`; the tables below are generated from those files by `code/experiments/make_tables.py`. Every reported bound includes a charge for the codec designs we tried on the same data during development (4 bits per description, and the disagreement term at confidence $\delta_2/16$; `code/experiments/accounting.py`).

5.1 Boolean tasks with a known input distribution

Setup. Inputs are uniform on $\{\pm 1\}^{50}$. Targets: the parity of 3 fixed coordinates (`parity3`), the majority of 5 (`maj5`), the AND of 3 (`and3`), a random truth table on 6 coordinates (`junta6`), and the majority of all 50 (`fullmaj`), plus `maj5` and `parity3` with 10% of the training labels flipped. The network f is an MLP of widths $(50, 512, 512, 1)$ with 289,281 parameters, trained with Adam (learning rate 10^{-3}, batch 64) until it reaches zero training error and a mean training loss below 0.01. We use $m \in \{250, \ldots, 16000\}$ and three seeds per setting (training set and initialization). The bound is Theorem 2 with the codec of Section 4; the disagreement term uses 2×10^5 fresh uniform inputs.

Results. Figure 1 and Table 1 show the main finding: wherever f generalizes, the bound on f is close to its test error. Across the learning curve the bound is roughly test error plus $L \ln 2/m$, where L is 28–100 bits (a truth table on the relevant coordinates or a linear threshold function). The parameter-counting baseline (32 bits per parameter, about 9.3×10^6 bits) and the VC baseline are 1.0 for every network. Where f has not learned the target (`parity3` at $m \leq 1000$), the bound is vacuous, correctly. Where f has partially learned it, the bound still tracks: at $m = 2000$, `parity3` networks have test error 0.29 and bound 0.30. The best g is then the true parity, with zero training errors, and $\Pr[f \neq g]$ is f's error.

Interpolating noisy labels. With 10% label noise, f fits every flipped label (zero training error). Its test error on noisy labels is $0.1 + 0.8 \Pr[f \neq \text{target}]$. The compressed g is the clean rule, so $\text{err}(g) \approx 0.1$ and $\Pr[f \neq g]$ measures the damage done by memorization. At $m = 16000$ the bounds are 0.17 against a test error of 0.14 for both tasks. A bound of this kind needs only that f is close to a short program on most of the input space; it does not need f to be simple.

5.2 Distribution-free bounds by exact equivalence

With $n = 20$ the whole input space has $2^{20} \approx 10^6$ points, so the disagreement between f and g can be computed exactly, and Corollary 3 turns compression into a bound that holds for *every* input distribution. Table 2 uses the same architecture and training protocol as before, with widths $(20, 512, 512, 1)$. Once a network has learned its target, its compressed description typically agrees with it on all 2^{20} inputs: the exception set E is empty and the distribution-free bound costs only the extra flag bit and the bits for $|E| = 0$ over the uniform-distribution bound. Where f has not learned the target, E is large, appending it is expensive, and the distribution-free bound is vacuous. That is correct here, since such an f could be wrong on most of the support of some distribution.

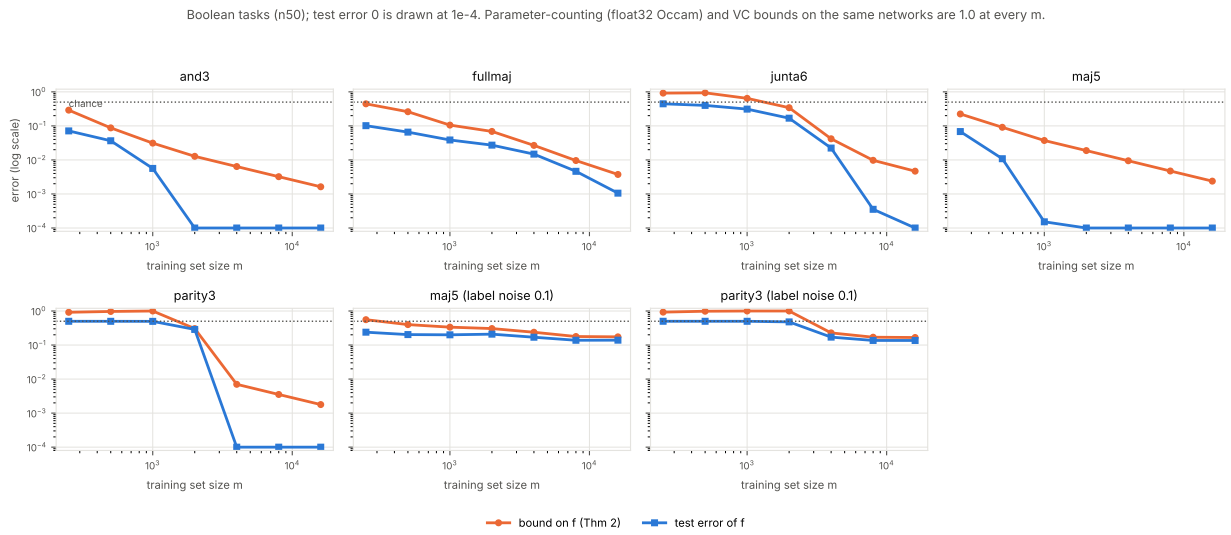


Figure 1: Boolean tasks, $n = 50$: test error of the trained network f (blue) and the certified upper bound on it from Theorem 2 (orange), averaged over 3 seeds, versus the number of training examples. The dotted line is chance.

5.3 Images: bounds with unlabeled inputs

With unlabeled inputs from D_X, Theorem 2 applies to image classifiers directly; this is the setting of Bazinet et al. [3], and we include it to place the other results. The surrogate g is the init-relative, DCT-coded, quantized network of Section 4, fine-tuned with a rate penalty using training labels, f's outputs, or both. The disagreement is measured on the 10,000 test images without their labels, with a union bound over the K compression configurations tried for each f. **??** reports the best Occam bound on g itself and the best bound on f. Our compressor is simple and uses a data-independent prior, so these bounds are much weaker than those of Bazinet et al. [3], who report 1.8–3.5% for an MNIST CNN using surrogates trained on part of the training set (data-dependent priors and sample compression). Coding the first-layer update in the DCT basis instead of the pixel basis matters a great deal here as well.

(table pending)

5.4 Images and Boolean tasks: bounds from training data alone

These are the bounds the question asks for in its strictest form: a deterministic bound on the error of the trained network itself, computed from the training set alone, with no validation data, no unlabeled data and no knowledge of the input distribution.

Worst-case weight sets (Theorem 4) are hopeless for dense networks. On a 784–100–10 MNIST network (test error 2.4%) we quantized the first-layer update in the DCT basis and certified with per-unit, per-band ℓ_2 balls around the quantized network that contain f. At every step size the certified first-layer radius is 12–13 times the actual first-layer error $|Rx|$. Certifying 95% of the training points needs steps of 2% of the update's RMS, i.e. 6.1×10^5 bits for the first layer alone, ten times more than $m = 6 \times 10^4$ (`results/mnist/t3_worstcase_probe.txt`).

Dithered compression (Theorem 5 and Lemma 7). Table 3 reports the bound for one-hidden-layer networks trained in the standard way (Adam, 20 epochs). It is non-vacuous (below the chance level of 0.9) for narrow networks: 0.44 for $H = 8$ (test error 7.0%) and 0.55 for $H = 16$ (5.1%) on MNIST, and 0.66 for $H = 16$ on Fashion-MNIST. It degrades with width: 0.70 at $H = 32$, 0.87 at $H = 64$, 0.97 at $H = 128$. Dithering does what it was designed to do. At comparable certified fractions it needs about 3 bits per first-layer coefficient instead of the 6–8 bits of worst-case sets (`results/mnist/t3_worstcase_probe.txt`), because the certificate scales with the ℓ_2 aggregate of the noise rather than its ℓ_1 worst case. But at the measured 2.1–3.6 bits per first-layer coefficient (3.2–3.6 for $H \leq 32$), $784H$ coefficients exceed $m = 6 \times 10^4$ bits once $H \gtrsim 25$, and the total grows linearly in H.

Three ablations on the 32-unit network: coding the first layer in the pixel basis instead of the DCT basis costs 36% more bits (bound 0.81 instead of 0.70); training with Gaussian weight noise of one or two times the initial RMS changes little (0.71, 0.69); a deterministic DCT initialization gives a more compressible network (67k instead of 84k bits; bound 0.69).

Networks with sparse updates. The cost is driven by the number of first-layer coefficients that have to be described. Since exactly-zero coefficients are sent exactly (no dither, no noise, Section 4), a network whose update is sparse in the DCT basis is much cheaper. We trained 100- and 256-unit networks normally, kept the largest 0.1–3% of the update's DCT coefficients and fine-tuned with that support fixed. Pruning followed by fine-tuning is a standard way to obtain small deployable networks [40], and the bound is still on the network that was trained and deployed. Table 4 shows the result. A standard-width MNIST network (100 hidden units) with 1% of its update coefficients nonzero has test error 4.6% and a training-data-only bound of 0.31. Keeping 0.3% gives 7.2% and 0.30, and 256-unit networks reach 0.40. The description is then 11–32 thousand bits. Each kept first-layer coefficient costs 5–8 bits for its value and 6–9 bits for its position, and the output layer takes 20–75% of the total. For the output layer the search always picked the largest step multiplier on its grid, so a wider grid would probably help a little.

Boolean networks: a bound for every input distribution. **??** applies the same theorem to one-hidden-layer networks on Boolean tasks. Since it uses only the training sample, the bound holds whatever distribution generated the inputs. With the uniform distribution known, Theorem 2 on the same networks is one to two orders of magnitude tighter, which measures what the known input distribution buys.

(table pending)

5.5 Initialization

Coding relative to the initialization. Table 5 trains the same 784–256–10 network from three initialization scales and from a deterministic DCT initialization. It then codes the update either relative to the (free) initialization or relative to zero, with the unlabeled-data codec of Section 5.3. The dithered training-data-only code is compared the same way in Table 3, on a 32-unit network.

The init-kernel codec. **??** compares, on the same Boolean networks ($n = 30$, one hidden layer of 512 units), the init-kernel codec of Proposition 10 with function-space distillation. The lazy regime uses initialization scale 2 and plain SGD; the rich regime uses scale 0.1 and Adam. The init-kernel codec describes $\theta_0 + P_r\Delta$, where P_r projects onto the top-r eigenvectors of the uniform-input empirical kernel at θ_0, estimated from 4000 uniform inputs; it is free given θ_0. Its candidate values

of r are the dimensions of the low-degree Walsh spaces, $1+n=31$, $1+n+\binom{n}{2}=466$ and 4526, together with powers of two.

(table pending)

6 Discussion

When is the reduction tight? Optimizing Theorem 2 over descriptions gives, up to estimation error,

$$\mathrm{err}(f) \;\lesssim\; \min_{\sigma}\Big[\mathrm{err}(g_\sigma)+\Pr_{D_X}[f\neq g_\sigma]+\tfrac{L(\sigma)\ln 2}{m}\Big] \;\le\; \mathrm{err}(f)+\epsilon+\tfrac{\ln 2}{m}\,K_\epsilon(f),$$

where $K_\epsilon(f)=\min\{L(\sigma):\Pr_{D_X}[f\neq g_\sigma]\le\epsilon\}$ is the ϵ-approximate description length of f under D_X (this rough form ignores the variance factor in the binomial inversion, which only helps at small errors). So the reduction is as good as f's *lossy* compressibility under the input distribution. Lossy is essential. A network that interpolates noisy labels differs from the clean target only near the training points, a set of small D_X-measure, so it can have small K_ϵ. Our label-noise experiments show the bound tracking the test error of such interpolating networks (Section 5.1). The bound fails, correctly, when f is far from every short program; this includes the case of a good network for an incompressible target. Generalization of a learning *algorithm* can always be reduced to compression in the other sense, that of the sample (Littlestone–Warmuth; Moran and Yehudayoff [28]), with the training procedure as decoder. This is a second description language for the same reduction, and its decoder can use the initialization for free.

Provable closeness without inputs from D_X. The second term of Theorem 2 is where a validation-free bound has to work. With known or sampleable D_X it is free; with unlabeled inputs it is the method of Bazinet et al. [3]. With neither, the only information about D_X is the training inputs, which f depends on. The dithered transfer theorem resolves this by making the certificate a function of the description alone. It costs a weight-space description of f precise enough for f's margins, and the cost grows with the number of weights that matter at typical inputs. Functional simplicity of f (e.g. f is close to a 5-junta) is invisible to it, unlike to Theorem 2. Closing this gap is the main open problem the experiments point to. A certificate would have to establish $f(x)=g(x)$ at a point from a description of g plus a short proof about f, as in the compact-proofs programme of Gross et al. [14], rather than from weight-space proximity.

The initialization. The worry that the random initialization must be described is unfounded at the level of validity (Lemma 9). Any data-independent randomness can be used by the code for free, and a seed costs 32 bits anyway. What remains is a question about the update. (i) Function-space codecs (our Boolean truth tables and distilled students) never refer to f's initialization, so it cannot cost anything. (ii) Weight-space codecs, which training-data-only bounds need, code the update relative to θ_0. Weights that receive no gradient (first-layer weights on pixels that are zero in every training image) have update exactly zero and cost almost nothing, whereas coding them relative to zero costs their full entropy. (iii) Exact zero initialization is a degenerate critical point for ReLU MLPs and does not train. A deterministic structured initialization trains and costs nothing, but a seeded random one also costs nothing. (iv) From a typical initialization, wide networks trained by gradient descent approach a deterministic limit (the kernel or the mean-field limit), so the initialization only adds fluctuations of order $\mathrm{width}^{-1/2}$. A function-space codec absorbs these into the disagreement term; a weight-space codec must pay for them or certify through them. In the linearized regime the free eigenbasis of the initial kernel gives a codec (Proposition 10) whose cost

reproduces the n^k sample complexity of kernel methods on degree-k targets. The rich regime, in which the network finds the relevant coordinates, is what makes trained networks short to describe.

Limitations. The training-data-only bound is proved for one-hidden-layer networks. The same argument extends layer by layer (pre-activation errors stay sub-Gaussian, though no longer independent across units), but we have not carried it out. Our codecs and step grids were designed with the datasets in mind, as in all of this literature: strictly, the code must be fixed before the sample is drawn. The Boolean experiments use small input dimension and synthetic targets. On images we bound MLPs, not convolutional networks, and our unlabeled-data bounds are far from the best published ones [3], which use data-dependent priors and stronger compressors.

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A Experimental details

Boolean tasks. Relevant coordinates are a fixed random subset (seed 1000). Training inputs, labels and label noise use a seed determined by the run's seed and m. The network has PyTorch's default initialization (uniform on $\pm 1/\sqrt{\text{fan-in}}$ for weights and biases), drawn from an explicit generator seeded by the run's seed. Training stops at zero training error and mean training loss at most 0.01, or after 4×10^4 steps. The influence estimates use 2000 inputs per coordinate. Students are distilled for 2000 Adam steps (batch 256, learning rate 3×10^{-3}, cosine schedule) on a pool of 10^5 uniform inputs labelled by f. Quantization-aware fine-tuning runs 500 steps at learning rate 10^{-3} for each of three steps $\{0.5, 0.25, 0.125\}\times$RMS per layer. Candidate selection uses 3×10^4 further uniform inputs. For $n = 50$, the certified disagreement uses 2×10^5 fresh inputs from a generator seeded with OS entropy after g is fixed. For $n = 20$ it is computed on all 2^{20} inputs. The predictor f whose error is bounded is the float64 evaluation of the trained weights.

Images. Pixels are scaled to $[0, 1]$. One-hidden-layer networks are trained with Adam (learning rate 10^{-3}, batch 128) for 20 epochs. Weight-noise training adds, at every step, Gaussian noise of the stated multiple of each tensor's initial RMS before the forward pass, and removes it before the update. Sparse-update networks are trained densely for 10 epochs in the parameterization $W = W_0 + AB^\top$. The largest coefficients of A are then kept and the network is fine-tuned for 5 epochs at learning rate 5×10^{-4} with the support fixed. The dither step grid has 162 points: a global scale $\gamma \in \{0.25, 0.5, 1, 2, 4, 8\}$ (first-layer steps $\gamma \cdot 10^{-2}$ times the band and unit factors), band exponents $p_b \in \{0.5, 0.75, 1\}$, unit exponents $p_u \in \{0, 0.5, 1\}$, and the bias and output-layer step $0.002\,\gamma\, v_m$ with $v_m \in \{1, 4, 16\}$. The point is chosen on 6000 training points with a selection dither. (A first pass used a fixed $v_m = 1$; its results are kept in `results/mnist/*search_v1.jsonl` and are not reported.)

Design charge. The codec families, step grids, band layout and certificate parameters were developed with exploratory runs on the same datasets. We therefore charge every description 4 bits, and take every disagreement term at confidence $\delta_2/16$. This covers a menu of up to 16 designs (`code/experiments/accounting.py`) and changes no reported number by more than a few units in the third decimal.

B Provenance and verification

This work was produced by an AI system (Claude) in one session. The literature review (`literature.md`) was assembled by AI agents from web-search results and from the public repositories of Bazinet et al. [3] and Lotfi et al. [24]. Partway through, the session's web-search budget was exhausted, so some citations are marked there as unconfirmed. The theory and its implementation were checked by four independent AI referees, each with a different lens: the proof of Theorem 5 and Lemma 7; code against proof, including an independent reimplementation of the certificate; the remaining theorems and the experimental protocols; and attempts to break the bounds numerically. Each claimed defect was then checked by an independent adjudicator. No error invalidating a theorem or

a bound was found. The defects found (statement scope, notation, test coverage, bookkeeping of selection effects) were fixed and are listed in `verification/README.md`. No human has checked the proofs or the code.

task	m	test err. f	bound on f	bits of g
and3	500	0.0366	0.0875	32
and3	1000	0.0056	0.0312	32
and3	2000	0.0000	0.0128	32
and3	4000	0.0000	0.0064	32
and3	16000	0.0000	0.0016	32
fullmaj	500	0.0656	0.2609	103
fullmaj	1000	0.0386	0.1056	77
fullmaj	2000	0.0272	0.0688	94
fullmaj	4000	0.0148	0.0267	60
fullmaj	16000	0.0011	0.0037	48
junta6	500	0.4005	0.9364	61
junta6	1000	0.3111	0.6451	78
junta6	2000	0.1685	0.3422	102
junta6	4000	0.0223	0.0418	102
junta6	16000	0.0000	0.0047	102
maj5	500	0.0109	0.0910	54
maj5	1000	0.0002	0.0373	49
maj5	2000	0.0001	0.0188	49
maj5	4000	0.0000	0.0095	49
maj5	16000	0.0000	0.0024	49
parity3	500	0.4982	0.9695	87
parity3	1000	0.4964	1.0000	225
parity3	2000	0.2876	0.3043	35
parity3	4000	0.0000	0.0070	35
parity3	16000	0.0000	0.0018	35
maj5, noise 0.1	500	0.2028	0.3987	60
maj5, noise 0.1	1000	0.1997	0.3352	60
maj5, noise 0.1	2000	0.2088	0.3050	53
maj5, noise 0.1	4000	0.1696	0.2376	60
maj5, noise 0.1	16000	0.1390	0.1741	53
parity3, noise 0.1	500	0.4991	0.9848	155
parity3, noise 0.1	1000	0.4987	1.0000	218
parity3, noise 0.1	2000	0.4786	1.0000	228
parity3, noise 0.1	4000	0.1708	0.2275	35
parity3, noise 0.1	16000	0.1362	0.1665	35

Table 1: Boolean tasks, $n = 50$: test error of the trained network f (289,281 parameters) and the bound on it (Theorem 2), averaged over 3 seeds; 'bits' is the description length of the compressed g (including the 4-bit design charge). The parameter-counting and VC bounds for f are 1.0 in every row.

task	m	test err. f	bound on f	bits of g	any-D bound	$\lvert E\rvert$
`fullmaj`	4000	0.0000	0.0078	41	0.0236	5
`junta6`	4000	0.0000	0.0168	93	0.0184	0
`maj5`	4000	0.0000	0.0080	42	0.0083	0
`parity3`	4000	0.0000	0.0061	31	0.0065	0

Table 2: Boolean tasks, $n = 20$: test error of the trained network f (273,921 parameters) and the bound on it (Theorem 2), averaged over 3 seeds; 'bits' is the description length of the compressed g (including the 4-bit design charge). The any-D bound (Corollary 3) holds for every input distribution; it patches g on the set E of inputs where $g \neq f$ (enumerated over all 2^{20} inputs). The parameter-counting and VC bounds for f are 1.0 in every row.

data	H	training	code	test err. f	bits	certified	bound on f	32-bit Occam
mnist	8	standard	DCT	0.0697	23490	0.897	0.439	0.94
mnist	16	standard	DCT	0.0512	43658	0.919	0.553	0.99
mnist	32	standard	DCT	0.0323	83976	0.949	0.704	1.00
mnist	32	standard	DCT, DCT init	0.0331	67169	0.903	0.691	1.00
mnist	32	standard	DCT, ref. 0	0.0323	84269	0.950	0.704	1.00
mnist	32	standard	pixel	0.0323	114699	0.941	0.805	1.00
mnist	64	standard	DCT	0.0250	157051	0.964	0.871	1.00
mnist	128	standard	DCT	0.0213	223592	0.863	0.970	1.00
mnist	16	noise 1	DCT	0.0473	51751	0.938	0.573	0.99
mnist	32	noise 1	DCT	0.0320	81281	0.937	0.708	1.00
mnist	64	noise 1	DCT	0.0218	147536	0.960	0.858	1.00
mnist	100	noise 1	DCT	0.0210	221482	0.971	0.939	1.00
mnist	32	noise 2	DCT	0.0319	75306	0.932	0.691	1.00
mnist	100	noise 2	DCT	0.0199	212872	0.969	0.933	1.00
fashion	16	standard	DCT	0.1407	43032	0.826	0.659	1.00
fashion	32	standard	DCT	0.1305	77000	0.831	0.793	1.00
fashion	64	standard	DCT	0.1245	149239	0.833	0.926	1.00
fashion	32	noise 1	DCT	0.1305	78906	0.811	0.813	1.00

Table 3: Bounds on the original one-hidden-layer network f from the 60,000 training examples alone (Theorem 5 and Lemma 7): no validation set and no unlabeled data. 'Certified': fraction of training points with $\pi_g = 1$. 'Noise σ': f trained with Gaussian weight noise of σ times each tensor's initial RMS. Chance level is 0.9.

data	H	kept coefficients	test err. f	bits	certified	bound on f	32-bit Occam
mnist	100	3.0%	0.0333	32455	0.963	0.407	1.00
mnist	100	1.0%	0.0462	17664	0.945	0.314	1.00
mnist	100	0.3%	0.0722	11423	0.909	0.304	1.00
mnist	256	0.3%	0.0441	23244	0.924	0.399	1.00
mnist	256	0.1%	0.0647	21673	0.916	0.398	1.00
fashion	100	1.0%	0.1693	20176	0.828	0.501	1.00
fashion	100	0.3%	0.1926	13170	0.798	0.468	1.00

Table 4: As Table 3, for networks whose first-layer update is exactly sparse in the DCT basis (trained, pruned to the stated fraction of coefficients, fine-tuned with the support fixed). The support is part of the description; only nonzero coefficients are dithered.

initialization	scale	code relative to	test err. f	bound on g	bits	bound on f	bits
seeded random	0.3	initialization	0.0199	0.384	34421	0.407	34421
seeded random	1	initialization	0.0197	0.367	32264	0.391	32264
seeded random	1	zero	0.0197	0.377	34095	0.399	34095

Table 5: MNIST, 784-256-10: description length and bounds when the update is coded relative to the (free) initialization or relative to zero, for three initialization scales and a deterministic initialization. Best of 8 compression configurations each.